POLITECNICO
MILANO 1863

Visual reconstruction of played music

Ahmed Hassan, ahmedhassan.hassan@mail.polimi.it
Shuvro Ahmed, shuvro.ahmed@mail.polimi.it
Umme Humaira Himi, ummehumaira.himi@mail.polimi.it

February 8, 2026

I. Problem Statement

The objective of this project is to analyze a video of a played piano keyboard in order to identify the pressed keys at each instant. Given a monocular RGB video, the task is decomposed into multiple stages, including keyboard localization, identifying individual keys, tracking fingertip positions, and determining which keys are pressed over time.

The system is required to operate on standard video without any specialized sensors. All detections must be expressed in the full image coordinate system to allow interaction between different processing stages.

The main challenge arises from the presence of hands, reflections, and background elements, which make direct analysis on the full image difficult. For this reason, the processing pipeline is structured into multiple stages, where intermediate representations are used to simplify subsequent steps.

II. Method

A. Overall Pipeline

B. Frame Selection and Preprocessing

C. Piano Key Detection

D. Fingertip Detection

E. Fingertip-to-Key Association

F. Pressed-Key Detection

III. Implementation

A. Frame Extraction and ROI Localization

B. White and Black Key Localization

C. Fingertip Detection

D. Key Mapping and Pressed-Key Logic

IV. Experiments and Results

A. Dataset

B. Qualitative Results

C. Quantitative Evaluation

D. Failure Cases and Limitations